Domain 9: Infrastructure Vulnerabilities

Operational Context

Smart grid agents balance load and generation in real-time. **FR1 = Maintain grid frequency at 60Hz. FR2 = Prevent equipment damage (Overload Protection).**

The Goal Hijacking Attack

Attackers use "Sensor Masquerading" to hijack the "Load Shedding" FR.

- **Mechanism**: A compromised IoT botnet injects "High Load" telemetry into the grid controller, while simultaneously suppressing "Generation" readings.
- **Hijack**: The agent is forced to execute its "Emergency Load Shedding" FR to "save" the grid from a phantom overload. The goal "Prevent Damage" is weaponized to cause a blackout.

OODA Loop Transients

1. **Observe**: Agent sees “Load $>$ Capacity” (False Data).
2. **Orient**: Emergency State triggered.
3. **Decide**: Cut power to Sector 7 (Hospital District).
4. **Act**: Blackout occurs; the grid was stable, but the *agent* was destabilized.

CIF Defense: Physics-Informed OODA

CIF integrates **Physics-Informed Neural Networks (PINNs)** into the Orientation phase.

- **Conservation of Energy**: The agent checks if the reported "High Load" is physically consistent with the current flow at the substations (Kirchhoff's Laws). If $\sum I_{in} \neq \sum I_{out}$ beyond noise margins, the sensor data is rejected.
- **Slow-Transient Filter**: The "Emergency Shed" FR has a built-in temporal damper. It requires the overload condition to persist for $> \Delta t$ (defined by thermal limits) before acting, filtering out fast, synthetic OODA transients that are characteristic of cyber-attacks.